Exercise 6

Exercise 6 — Company X, first-year balance sheet and partly paid capital

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Company X was incorporated on 1/01/2017 with a subscription of shares worth 100 m.u., fully paid in cash. On 1/01/2017 it bought, with cash: a) an area (land) for 20 m.u.; b) equipment with a life of 10 years (straight-line depreciation) and a purchase price of 20 m.u.; c) raw materials for 20 m.u.

The company hires an employee for a total annual cost (salaries, Social Security, etc.) of 10 m.u. per year. This employee processes half of the raw material and sells it for a total of 50 m.u. Sales are the same every day of the year. The average collection period is 3 months.

The single tax rate on the profits of the company is 30% (paid on 31/12/2017). The company follows a strict self-financing strategy.

1. *Prepare the balance sheet at 31/12/2017, justifying and explaining the intermediate calculations.*
2. *Explain how the accounting statements would change if, instead of 100%, the shareholders had paid only 90% of the 100 m.u. subscribed.*

Phase 1 — Setup: data and hypotheses

Period: calendar year 2017. **Closing:** 31/12/2017.

Hypotheses:

1. The employee's cost is paid during the year.
2. Land has no useful life given and is not depreciated; the equipment has no residual value given (zero).
3. **Strict self-financing** [Slide 23]: the whole net income is retained (pay-out 0%).
4. The tax is paid on 31/12, so no tax liability remains.

Phase 2 — Frames: classification of every operation

Operation	Income statement 2017	Cash 2017	Open at 31/12/2017
Capital	–	+100	Share capital 100
Land	–	-20	Land 20
Equipment	Depreciation 2	-20	Equipment 20 – 2
Raw materials	COGS 10 (half consumed)	-20	Inventory 10
Employee	Personnel 10	-10	–

Operation	Income statement 2017	Cash 2017	Open at 31/12/2017
Sales	Revenue 50	+37.5	Receivables 12.5
Tax	8.4	−8.4	–

Phase 3 — Derivation (requirement 1)

Step 1. Depreciation and cost of goods sold

$$D = \frac{20 - 0}{10} = 2 \text{ m.u.}, \qquad \text{COGS} = \frac{20}{2} = 10 \text{ m.u.}, \qquad \text{Inventory} = 20 - 10 = 10 \text{ m.u.}$$

Step 2. Receivables from the collection period

Sales are spread evenly over the year and customers pay 3 months after buying. At 31/12 the sales of the last 3 months are still unpaid:

$$\text{Receivables} = 50 \times \frac{3}{12} = 12.5 \text{ m.u.}, \qquad \text{Collected} = 50 - 12.5 = 37.5 \text{ m.u.}$$

Step 3. Income statement 2017

Line	Computation	m.u.
Sales revenue		50.0
− Cost of goods sold	Step 1	10.0
= Gross profit		**40.0**
− Personnel		10.0
− Depreciation	Step 1	2.0
= Income from operations = income before taxes	no interest	**28.0**
− Income taxes (30%)	0.30×28	8.4
= Net income	$28 - 8.4$	**19.6**

All 19.6 is retained (strict self-financing).

Step 4. Cash at 31/12/2017

$$\text{Cash} = 100 - 20 - 20 - 20 + 37.5 - 10 - 8.4 = 59.1 \text{ m.u.}$$

Step 5. Balance sheet at 31/12/2017

Assets	m.u.	Liabilities and equity	m.u.
Land	20.0	Share capital	100.0
Equipment (20 − 2)	18.0	Retained earnings (net income 2017)	19.6
Non-current assets	**38.0**	**Equity**	**119.6**
Inventory (raw materials)	10.0	**Liabilities**	**0**
Accounts receivable	12.5		
Cash	59.1		
Current assets	**81.6**		
Total assets	**119.6**	**Total liabilities and equity**	**119.6**

Phase 3b — Derivation (requirement 2: only 90% paid)

Step 6. What changes

The shareholders **subscribed** 100 m.u. but **paid in** only $0.90 \times 100 = 90$ m.u. The remaining 10 m.u. is capital not yet paid (uncalled capital): the shareholders still owe it to the company.

- **Income statement: no change**. Revenues and expenses do not depend on how the capital was paid; there is no interest involved. Net income stays 19.6.
- **Cash:** 10 m.u. less: $59.1 - 10 = 49.1$ m.u. All the purchases (60) can still be paid in cash, since the company receives 90.
- **Equity:** the contributed capital is what the stockholders actually provided [Slide 13, Slide 17], so the subscribed capital is shown minus the unpaid part: $100 - 10 = 90$.

Assets	m.u.	Liabilities and equity	m.u.
Non-current assets	**38.0**	Subscribed capital	100.0
Inventory	10.0	– Uncalled capital (unpaid by shareholders)	(10.0)
Accounts receivable	12.5	Retained earnings	19.6
Cash	49.1	**Equity**	**109.6**
Current assets	**71.6**	**Liabilities**	**0**
Total assets	**109.6**	**Total liabilities and equity**	**109.6**

Both totals fall by exactly the 10 m.u. not paid.

Phase 4 — Interpretation and checks

- **Balance:** $38 + 81.6 = 119.6 = 100 + 19.6$ (requirement 1); $38 + 71.6 = 109.6 = 90 + 19.6$ (requirement 2).
- **Equity bridge:** equity = capital paid + net income retained: $100 + 19.6$ and $90 + 19.6$.
- **Profit vs cash:** after the purchases of 1/01, the cash generated during the year is $37.5 - 10 - 8.4 = 19.1$, while the profit is 19.6. Reconciliation: $19.6 + 2 + 10 - 12.5 = 19.1$, that is, profit + depreciation (expense, never a payment) + materials consumed (expense whose payment was made on 1/01 with the purchase) − sales not collected yet.
- **Presentation note:** some textbooks show the 10 m.u. owed by shareholders as an asset instead of deducting it from capital; then both totals stay at 119.6. The deduction used here is the one consistent with "contributed capital" in [Slide 17]. Both readings give the same net equity provided (109.6) and leave the income statement unchanged.

Check against the official key

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Requirement 1 coincides with the key: sales 50; cost of goods sold 10; personnel 10; depreciation 2; EBIT 28; tax 8.4; net income 19.6; cash 59.1; receivables 12.5; inventory 10; land 20; equipment 18; total assets 119.6. The key does not solve requirement 2 (90 % paid in), so that part of the solution is not covered by it.